

An Analysis of Lasso regression

Lasso regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|_1$$

Lasso regression -- Part 1

j lasso .
$$\hat{\beta}_j = \frac{\hat{\beta}_{OLS}}{\sqrt{1 + \lambda_j}} \max\left(0, 1 - \frac{\lambda_j}{\|\hat{\beta}_{OLS}\|_1}\right) = \frac{\hat{\beta}_{OLS}}{1 + \lambda_j} \max\left(0, 1 - \frac{\lambda_j}{\|\hat{\beta}_{OLS}\|_1}\right)$$

Lasso regression -- Part 2

where only β is penalized while α is free to take any allowed value, just as β_0 was not penalized in the basic case.

Lasso regression -- Part 3

Group lasso In 2006, Yuan and Lin introduced the group lasso to allow predefined groups of covariates to jointly be selected into or out of a model. This is useful in many settings, perhaps most obviously when a categorical variable is coded as a collection of binary covariates. In this case, group lasso can ensure that all the variables encoding the categorical covariate are included or excluded together. Another setting in which grouping is natural is in biological studies. Since genes and proteins often lie in known pathways, which pathways are related to an outcome may be more significant than whether individual genes are. The objective function for the group lasso is a natural generalization of the standard lasso objective

Lasso regression -- Part 4

The loss function of the lasso is not differentiable, but a wide variety of techniques from convex analysis and optimization theory have been developed to compute the solutions path of the lasso. These include coordinate descent, subgradient methods, least-angle regression (LARS), and proximal gradient methods. Subgradient methods are the natural generalization of traditional methods such as gradient descent and stochastic gradient descent to the case in which the objective function is not differentiable at all points. LARS is a method that is closely tied to lasso models, and in many cases allows them to be fit efficiently, though it may not perform well in all circumstances. LARS generates complete solution paths. Proximal methods have become popular because

of their flexibility and performance and are an area of active research. The choice of method will depend on the particular lasso variant, the data and the available resources. However, proximal methods generally perform well. The "glmnet" package in R, where "glm" is a reference to "generalized linear models" and "net" refers to the "net" from "elastic net" provides an extremely efficient way to implement LASSO and some of its variants. The "celer" package in Python provides a highly efficient solver for the Lasso problem, often outperforming traditional solvers like scikit-learn by up to 100 times in certain scenarios, particularly with high-dimensional datasets. This package leverages dual extrapolation techniques to achieve its performance gains. The celer package is available at GitHub.

Lasso regression -- Part 5

j OLS =
$$(X^T X)^{-1} X^T y = X^+ y$$

Lasso regression -- Part 6

where $\|\cdot\|_0$ is the " ℓ_0 norm", which is defined as $\|z\|_0 = m$ if exactly m components of z are nonzero. Again assuming orthonormal covariates, it can be shown that in this special case

Lasso regression -- Part 7

$$\min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$$

Lasso regression -- Part 8

That is, if regressors are uncorrelated, λ again specifies the minimal influence of β_0 . Even when regressors are correlated, the first time that a regression parameter is activated occurs when λ is equal to the highest diagonal element of $R \otimes I$. These results can be compared to a rescaled version of the lasso by defining $q_{lasso, i} = \frac{1}{\sum_{l=1}^p |b_{OLS, l} - \beta_0|}$, which is the average absolute deviation of b_{OLS} from β_0 . Assuming that regressors are uncorrelated, then the moment of activation of the i^{th} regressor is given by

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Lasso regression -- Part 9

$$\min_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \right\},$$
$$\left\{ \min_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \right\} \right\}$$

Lasso regression -- Part 10

$$b_{\ell_1} = \begin{cases} (1 - \lambda / R^2) b_{\text{OLS}} & \text{if } \lambda \leq R^2 \\ 0 & \text{if } \lambda > R^2 \end{cases}$$
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Lasso regression -- Part 11

$$\|\hat{\beta}_{\lambda}\|_1 \leq \sqrt{N\lambda} \quad \left\{ \|\hat{\beta}_{\lambda}\|_1 \leq \sqrt{N\lambda} \right\}$$
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Lasso regression -- Part 12

Lasso can set coefficients to zero, while the superficially similar ridge regression cannot. This is due to the difference in the shape of their constraint boundaries. Both lasso and ridge regression can be interpreted as minimizing the same objective function

Lasso regression -- Part 13

Ridge regression shrinks all coefficients by a uniform factor of $(1 + N\lambda)^{-1}$ and does not set any coefficients to zero. It can also be compared to regression with best subset selection, in which the goal is to minimize

Lasso regression -- Part 14

Generalizations Lasso variants have been created in order to remedy limitations of the original technique and to make the method more useful for particular problems. Almost all of these focus on respecting or exploiting dependencies among the covariates. Elastic net regularization adds an additional ridge regression-like penalty that improves performance when the number of predictors is larger than the sample size, allows the method to select strongly correlated variables together, and improves overall prediction accuracy. Group lasso allows groups of related covariates to be selected as a single unit, which can be useful in settings where it does not make sense to include some covariates without others. Further extensions of group lasso perform variable selection within individual groups (sparse group

lasso) and allow overlap between groups (overlap group lasso). Fused lasso can account for the spatial or temporal characteristics of a problem, resulting in estimates that better match system structure. Lasso-regularized models can be fit using techniques including subgradient methods, least-angle regression (LARS), and proximal gradient methods. Determining the optimal value for the regularization parameter is an important part of ensuring that the model performs well; it is typically chosen using cross-validation.